Strong Law of Large Numbers

Stats 220A

What the SLLN adds

WLLN: $S_n/n \xrightarrow{\mathbb{P}} \mu$. Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \varepsilon\right) \to 0.$$

This controls the *probability* of deviation at each n, but does not assert that a given sample path eventually stays close to μ.

SLLN: $S_n/n \xrightarrow{\text{a.s.}} \mu$. For *almost every* ω, the sequence $S_n(\omega)/n$ converges to μ.

The difference. WLLN allows the bad set to drift with n — different ω's can fail at different times. SLLN requires a single null set that covers all failures simultaneously.

The surprise. Etemadi (1981) proved the SLLN under *pairwise* independence. Full mutual independence is not needed.

Compare: the WLLN often follows under uncorrelatedness (via an L^2 argument). Etemadi's SLLN needs pairwise independence but not full independence.

Etemadi's SLLN: statement

Theorem (Etemadi 1981)

Let $X_1, X_2, \ldots$ be **pairwise independent** *and identically distributed with $\mathbb{E}|X_1| < \infty$. Let $\mu = \mathbb{E}X_1$ and $S_n = X_1 + \cdots + X_n$. Then*

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu.$$

Proof strategy. Three steps:

1. **Truncation.** Replace X_k by $Y_k = X_k \mathbf{1}_{\{|X_k| \le k\}}$. Show S_n and $\tilde{S}_n = \sum_k Y_k$ agree eventually a.s.
2. **Subsequence.** Prove $\tilde{S}_{n_k}/n_k \xrightarrow{\text{a.s.}} \mu$ along $n_k = \lceil \alpha^k \rceil$ for $\alpha > 1$ using a variance bound and Borel–Cantelli.
3. **Interpolation.** Fill in between subsequence values using monotonicity of partial sums (after passing to nonneg parts).

Each step is elementary; the cleverness is in choosing the right subsequence.

Step 1: truncation

Define $Y_k = X_k \mathbf{1}_{\{|X_k| \le k\}}$. The key fact:

Lemma

$\sum_{k=1}^{\infty} \mathbb{P}(X_k \neq Y_k) < \infty$.

Proof.

Since the X_k are i.i.d.,

$$\sum_{k=1}^{\infty} \mathbb{P}(|X_k| > k) = \sum_{k=1}^{\infty} \mathbb{P}(|X_1| > k).$$

Moreover,

$$\sum_{k=1}^{\infty} \mathbb{P}(|X_1| > k) \le \sum_{k=1}^{\infty} \int_{k-1}^{k} \mathbb{P}(|X_1| > t)\, dt = \int_0^{\infty} \mathbb{P}(|X_1| > t)\, dt = \mathbb{E}|X_1| < \infty.$$

Hence $\sum_k \mathbb{P}(X_k \neq Y_k) < \infty$, so by BC1, $\mathbb{P}(X_k \neq Y_k \text{ i.o.}) = 0$. Therefore there exists $K(\omega) < \infty$ such that for all $k \ge K(\omega)$, $X_k(\omega) = Y_k(\omega)$, and thus $\tilde{S}_n = \sum_{k=1}^{n} Y_k$ eventually agree a.s., and it suffices to show $\tilde{S}_n/n \xrightarrow{\text{a.s.}} \mu$. □

Note. This is the same truncation idea as in the WLLN proof, now applied to get an a.s. result via BC1 rather than a probabilistic bound.

A key variance summability bound (Durrett/Etemadi)

Lemma

Let $Y_k = X_k \mathbf{1}_{\{|X_k| \le k\}}$. *Then*

$$\sum_{k=1}^{\infty} \frac{\mathrm{Var}(Y_k)}{k^2} \;\le\; 4\,\mathbb{E}|X_1|.$$

Proof.

Since $\mathrm{Var}(Y_k) \le \mathbb{E}[Y_k^2]$ and $|Y_k| \le k$,

$$\mathbb{E}[Y_k^2] = \int_0^{\infty} 2y\,\mathbb{P}(|Y_k| > y)\,dy \le \int_0^{k} 2y\,\mathbb{P}(|X_1| > y)\,dy.$$

Therefore, by Tonelli/Fubini (nonnegative terms),

$$\sum_{k=1}^{\infty} \frac{\mathbb{E}[Y_k^2]}{k^2} \le \int_0^{\infty} \Big(\sum_{k>y} k^{-2}\Big)\, 2y\,\mathbb{P}(|X_1| > y)\,dy.$$

For all $y \ge 0$, one checks the crude bound $2y \sum_{k>y} k^{-2} \le 4$. Hence

$$\sum_{k=1}^{\infty} \frac{\mathrm{Var}(Y_k)}{k^2} \le \sum_{k=1}^{\infty} \frac{\mathbb{E}[Y_k^2]}{k^2} \le 4 \int_0^{\infty} \mathbb{P}(|X_1| > y)\,dy = 4\mathbb{E}|X_1|.$$

Step 2: a geometric subsequence + Borel–Cantelli

Fix $\alpha > 1$ and define $k(n) = \lfloor \alpha^n \rfloor$. Let $\tilde{S}_m = \sum_{j=1}^{m} Y_j$ and $\tilde{\mu}_m = \mathbb{E}[\tilde{S}_m]$.

Lemma

For every $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\left| \tilde{S}_{k(n)} - \tilde{\mu}_{k(n)} \right| > \varepsilon k(n) \right) < \infty,$$

hence

$$\frac{\tilde{S}_{k(n)} - \tilde{\mu}_{k(n)}}{k(n)} \xrightarrow{\text{a.s.}} 0.$$

Conclusion. Since $\mathbb{E} Y_m \to \mu$ and Cesàro averaging preserves limits,

$$\frac{\tilde{\mu}_{k(n)}}{k(n)} = \frac{1}{k(n)} \sum_{m=1}^{k(n)} \mathbb{E} Y_m \to \mu,$$

so $\tilde{S}_{k(n)}/k(n) \xrightarrow{\text{a.s.}} \mu$.

Proof of Lemma

Proof.

By Chebyshev and pairwise independence (variance additivity),

$$\mathbb{P}\Big(\Big|\tilde{S}_{k(n)} - \tilde{\mu}_{k(n)}\Big| > \varepsilon k(n)\Big) \le \frac{\mathrm{Var}(\tilde{S}_{k(n)})}{\varepsilon^2 k(n)^2} = \frac{1}{\varepsilon^2}\sum_{m=1}^{k(n)} \frac{\mathrm{Var}(Y_m)}{k(n)^2}.$$

Sum over n and swap sums (Tonelli):

$$\sum_{n=1}^{\infty} \mathbb{P}(\cdots) \le \frac{1}{\varepsilon^2}\sum_{m=1}^{\infty} \mathrm{Var}(Y_m) \sum_{n:\, k(n)\ge m} \frac{1}{k(n)^2}.$$

Since $k(n) \asymp \alpha^n$, the inner tail is a geometric series and satisfies

$$\sum_{n:\, k(n)\ge m} \frac{1}{k(n)^2} \lesssim \frac{C_\alpha}{m^2}.$$

Thus

$$\sum_{n=1}^{\infty} \mathbb{P}(\cdots) \lesssim \frac{C_\alpha}{\varepsilon^2}\sum_{m=1}^{\infty} \frac{\mathrm{Var}(Y_m)}{m^2} < \infty$$

so by the previous lemma. By BC1, we have the proof.

Step 3: interpolation

Reduce to the case $X_k \geq 0$ by writing $X_k = X_k^+ - X_k^-$ and applying the argument to (X_k^+) and (X_k^-) separately.

Fix $\alpha > 1$ and set $n_j = \lceil \alpha^j \rceil$ (equivalently, $n_j = k(j)$ up to rounding). From Step 2 we have $\tilde{S}_{n_j}/n_j \xrightarrow{\text{a.s.}} \mu$.
For $n_j \leq n < n_{j+1}$, since $\tilde{S}_n$ is nondecreasing,

$$\frac{\tilde{S}_{n_j}}{n_{j+1}} \leq \frac{\tilde{S}_n}{n} \leq \frac{\tilde{S}_{n_{j+1}}}{n_j}.$$

Since $n_{j+1}/n_j \to \alpha$, we obtain

$$\frac{1}{\alpha}\mu \leq \liminf_{n\to\infty} \frac{\tilde{S}_n}{n} \leq \limsup_{n\to\infty} \frac{\tilde{S}_n}{n} \leq \alpha\mu \quad \text{a.s.}$$

Letting $\alpha \downarrow 1$ along a countable sequence (so the exceptional null set stays null) yields $\tilde{S}_n/n \xrightarrow{\text{a.s.}} \mu$.

Why a geometric subsequence? It is sparse enough that the Borel–Cantelli sum in Step 2 is finite (via a geometric-series tail bound), but dense enough that consecutive terms satisfy $n_{j+1}/n_j \approx \alpha$, which makes the interpolation work.

Why we need a maximal inequality

In the SLLN proof above, we controlled S_n at a fixed n (or along a subsequence). But many arguments require controlling $\max_{k \le n} |S_k|$ simultaneously.

The naive bound is $\mathbb{P}(\max_{k \le n} |S_k| > x) \le \sum_{k=1}^{n} \mathbb{P}(|S_k| > x)$, but this loses a factor of n.

Kolmogorov's idea. For independent summands, we can do much better: the maximum partial sum satisfies the *same* type of bound as the terminal sum, up to a constant.

Theorem (Kolmogorov's Maximal Inequality)

Let $X_1, \ldots, X_n$ be independent with $\mathbb{E}X_k = 0$ and $\mathbb{E}X_k^2 < \infty$. Then for any $x > 0$:

$$\mathbb{P}\left(\max_{1 \le k \le n} |S_k| \ge x\right) \le \frac{\mathrm{Var}(S_n)}{x^2}.$$

The right side depends only on $\mathrm{Var}(S_n)$, not on $n \cdot \max_k \mathrm{Var}(S_k)$.

Proof of Kolmogorov's maximal inequality

Let $A = \{\max_{k \le n} |S_k| \ge x\}$ and $A_k = \{|S_1| < x, \dots, |S_{k-1}| < x, |S_k| \ge x\}$, the event that k is the *first* time $|S_k| \ge x$. The A_k are disjoint and $A = \bigcup_k A_k$.
On A_k, we have $|S_k| \ge x$. Write $S_n = S_k + (S_n - S_k)$, so $S_n^2 \ge S_k^2 + 2S_k(S_n - S_k)$. Integrating over A_k:

$$\int_{A_k} S_n^2 \, d\mathbb{P} \ge \int_{A_k} S_k^2 \, d\mathbb{P} + 2 \int_{A_k} S_k(S_n - S_k) \, d\mathbb{P}.$$

The second term vanishes: A_k depends only on $X_1, \dots, X_k$, and $S_n - S_k = X_{k+1} + \cdots + X_n$ is independent of $\mathcal{F}_k$ with mean zero, so $\mathbb{E}[\mathbf{1}_{A_k} S_k(S_n - S_k)] = \mathbb{E}[\mathbf{1}_{A_k} S_k] \cdot \mathbb{E}[S_n - S_k] = 0$.
Therefore $\int_{A_k} S_n^2 \, d\mathbb{P} \ge x^2 \mathbb{P}(A_k)$. Summing over k:

$$\mathrm{Var}(S_n) = \mathbb{E}[S_n^2] \ge \sum_{k=1}^{n} \int_{A_k} S_n^2 \, d\mathbb{P} \ge x^2 \mathbb{P}(A). \quad \square$$

Convergence of random series

Theorem (Kolmogorov)

Let $X_1, X_2, \ldots$ be independent with $\mathbb{E}X_k = 0$. If $\sum_{k=1}^{\infty} \mathbb{E}X_k^2 < \infty$, then $\sum_{k=1}^{\infty} X_k$ converges a.s.

Proof.

Let $S_n = \sum_{k=1}^{n} X_k$ and $S = \lim S_n$ if it exists. It suffices to show $\{S_n\}$ is a.s. Cauchy. For $m < n$, the maximal inequality applied to $X_{m+1}, \ldots, X_n$ gives:

$$\mathbb{P}\left(\max_{m<k\leq n} |S_k - S_m| \geq \varepsilon\right) \leq \frac{1}{\varepsilon^2} \sum_{k=m+1}^{n} \mathbb{E}X_k^2.$$

Since $\sum_k \mathbb{E}X_k^2 < \infty$, the right side $\to 0$ as $m \to \infty$ uniformly in n. Letting $n \to \infty$: $\mathbb{P}(\sup_{k>m} |S_k - S_m| \geq \varepsilon) \to 0$, so S_n is a.s. Cauchy, hence a.s. convergent. $\square$

Application. $\sum_{k=1}^{\infty} X_k/k$ converges a.s. whenever X_k are i.i.d. with $\mathbb{E}X_k = 0$ and $\mathbb{E}X_k^2 < \infty$ (since $\sum 1/k^2 < \infty$).

Application: Glivenko–Cantelli theorem

Definition

The **empirical CDF** of $X_1, \ldots, X_n$ i.i.d. with CDF F is $F_n(x) = \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{X_k \le x\}}$.

Theorem (Glivenko–Cantelli)

$\sup_{x\in\mathbb{R}} |F_n(x) - F(x)| \xrightarrow{\text{a.s.}} 0.$

Proof sketch.

For each fixed x, $F_n(x)$ is an average of i.i.d. Bernoulli$(F(x))$ variables, so $F_n(x) \xrightarrow{\text{a.s.}} F(x)$ by the SLLN.

The hard part is uniformity over x. Fix $\varepsilon > 0$ and choose a finite grid $-\infty = x_0 < x_1 < \cdots < x_m = \infty$ with $F(x_j) - F(x_{j-1}^-) \le \varepsilon$. Choose the grid points x_j to be continuity points of F (possible since a CDF has at most countably many discontinuities). The SLLN gives $F_n(x_j) \xrightarrow{\text{a.s.}} F(x_j)$ for each j. For $x \in [x_{j-1}, x_j]$, monotonicity of F_n and F controls $|F_n(x) - F(x)|$ by ε plus the error at the grid points. $\square$

Statistical meaning. The empirical CDF is a *uniformly consistent* estimator of the true CDF. The entire distribution function is recoverable from data, not just individual quantiles.

Summary

Result	**Key idea**
Etemadi's SLLN	Truncation + geometric subsequence + BC1 + interpolation
Kolmogorov maximal ineq.	Control $\max \lvert S_k \rvert$ via independence of increments
Convergence of series	Cauchy criterion + maximal inequality
Glivenko–Cantelli	SLLN pointwise + uniform control via finite grid

The independence hierarchy revisited.

- WLLN: uncorrelated suffices (L^2 approach).
- SLLN: pairwise independence suffices (Etemadi).
- Kolmogorov maximal ineq.: requires full independence of increments.